1. $r_e = 8.39\Omega$
   $Z_i = 503\Omega$
   $Z_o = 2{,}2k\Omega$
   $A_V = -262$; $A_i = 60$
2. $r_e = 32\Omega$
   $Z_i = 1{,}8k\Omega$; $Z_o = 3{,}9k\Omega$
   $A_v = -122$; $A_i = 56$
3. $r_e = 19\Omega$
   $Z_i = 3{,}18k\Omega$; $Z_o = 6{,}8k\Omega$
   $A_v = -357{,}89$; $A_i = 167{,}36$.
   DCLL: $I_C = -0.11V_{CE} + 2{,}22$ (mA)
   ACLL: $I_{C\sum} = -0{,}147V_{CE\sum} + 2{,}5$ (mA)
   Maxswing($i_{c(p-p)}$) = 2,26 (mA)
   R_{B1} =
4. $r_e = 5{,}34\Omega$
   $Z_i = 118{,}3k\Omega$; $Z_o = 2{,}2k\Omega$
   $A_v = -1{,}81$; $A_i = 97{,}39$
5. $r_e = 7{,}07\Omega$
   $Z_i = 74{,}7k\Omega$; $Z_o = 5{,}6k\Omega$
   $A_v = -4{,}64$; $A_i = 61{,}89$
6. $r_e = 8{,}72\Omega$
   $Z_i = 142{,}25k\Omega$; $Z_o = 8{,}69\Omega$
   $A_v = 0{,}997$; $A_i = -52{,}53$
7. $r_e = 28{,}05\Omega$
   $Z_i = 7{,}03k\Omega$; $Z_o = 27{,}66\Omega$
   $A_v = 0{,}986$; $A_i = -3{,}47$
8. $r_e = 33{,}38\Omega$
   $Z_i = 33{,}38\Omega$; $Z_o = 4{,}7k\Omega$
   $A_v = 140{,}8$; $A_i \approx 1$
9. $r_e = 23{,}636\Omega$
   $Z_i = 23{,}636\Omega$; $Z_o = 3{,}6k\Omega$
   $A_v = 163{,}2$; $A_i = 0{,}9868 \approx 1$
10. $r_e = 13{,}2\Omega$
    $Z_i = 503{,}46\Omega$; $Z_o = 3{,}83k\Omega$
    $A_v = -295{,}45$; $A_i = 38{,}1$
11. $r_e = 8{,}04\Omega$
    $Z_i = 0{,}62k\Omega$; $Z_o = 1{,}66k\Omega$
    $A_v = -206{,}45$; $A_i = 71{,}28$
12. $r_e = 6{,}14\Omega$
    $Z_i = 816{,}21\Omega$; $Z_o = 72{,}94k\Omega$
    $A_v = -357{,}68$; $A_i = 132{,}7$
    $A_{VS} = -160{,}74$

13. $r_e = 7{,}81\Omega$

$Z_i = 616{,}52\Omega$; $Z_o = 4{,}3k\Omega$

$A_{vNL} = -557{,}36$; $A_V = -214{,}98$; $A_i = 49{,}04$

$A_{VS} = -81{,}91$

$V_o = -429{,}96\sin 1000t$ (mV)

15. $g_m = 1{,}5$ mS

$Z_i = 1M\Omega$; $Z_o = 1{,}8k\Omega$

$A_v = -2{,}7$

16. $g_m = 1{,}2$mS

$Z_i = 10M\Omega$; $Z_o = 3{,}3k\Omega$

$A_v = -4$

17. $g_m = 5{,}4$ mS

$Z_i = 9{,}7\ M\Omega$; $Z_o = 2k\Omega$

$A_v = -10{,}8$

18. $g_m = 2{,}15$mS

$Z_i = 1M\Omega$; $Z_o = 2{,}7k\Omega$

$A_v = -3{,}69$; $A_{VS} = -3{,}69$